Perform Basic Data Transformation in ADF

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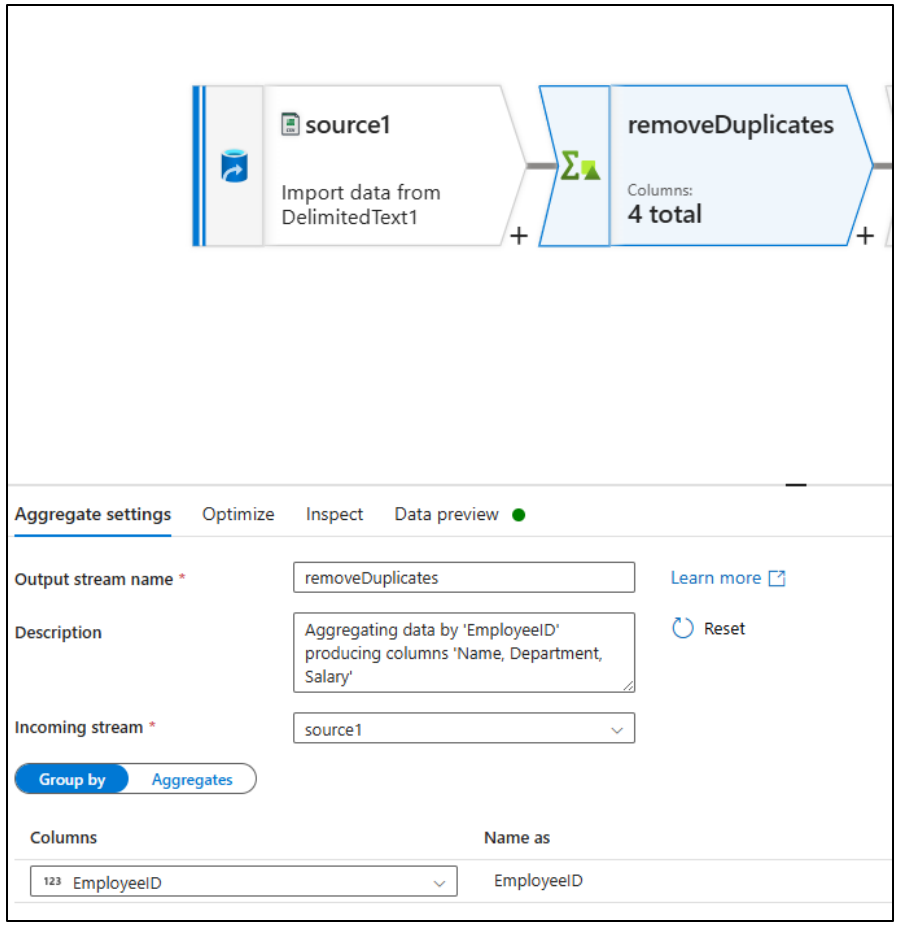
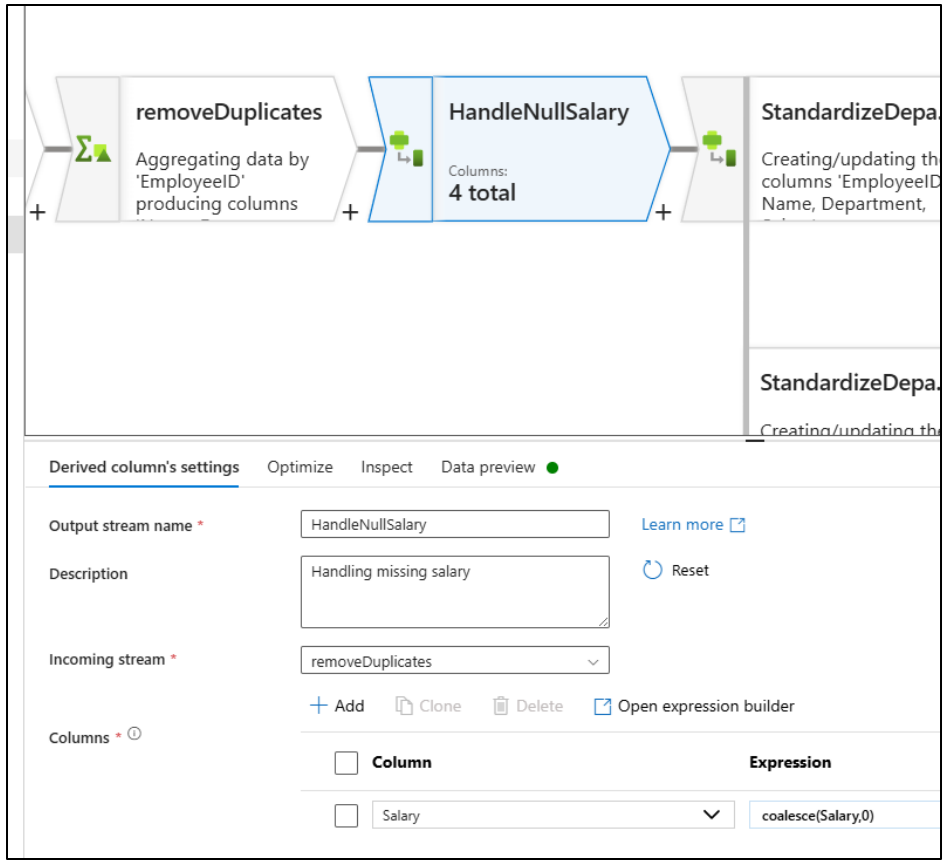
Dept: Data Engineering

1. **Data Cleaning**

a) Remove duplicate employee records.

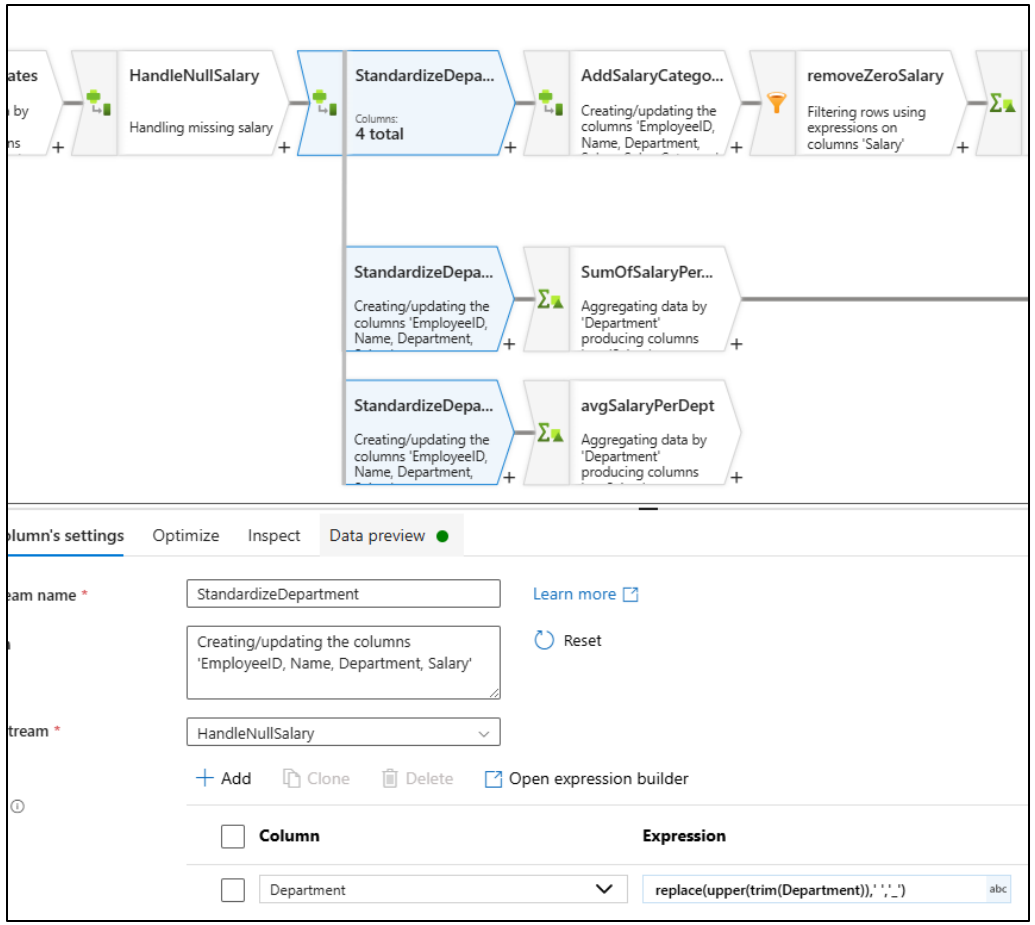
b) Handle missing or null salary values (e.g., replace with 0 or flag them).





2. Data Transformation

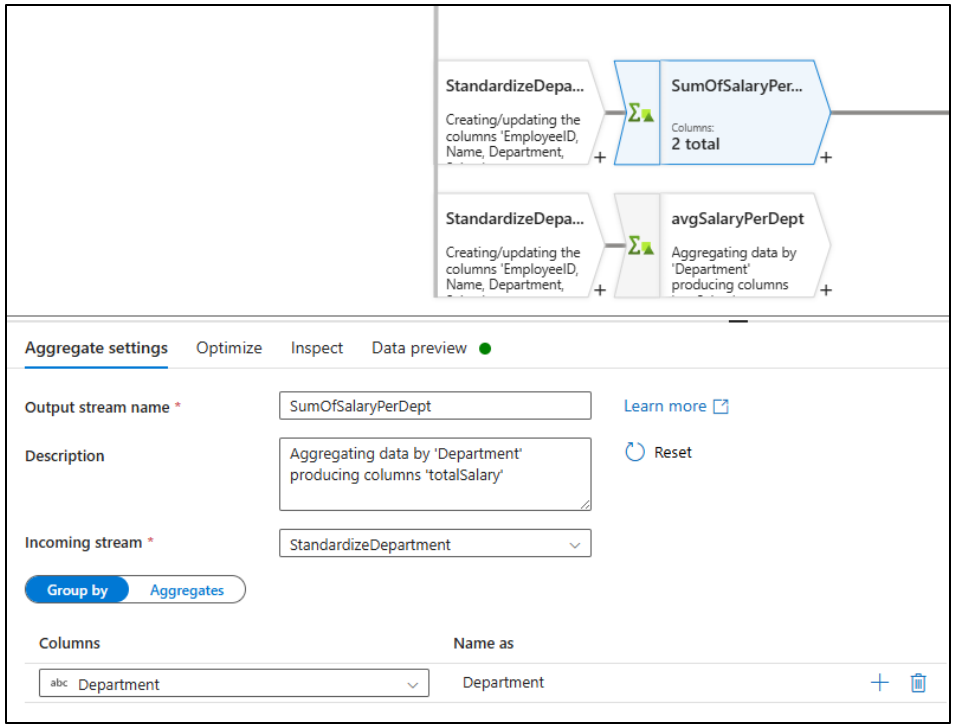
a) Standardize department names (convert to uppercase, replace spaces with underscores).

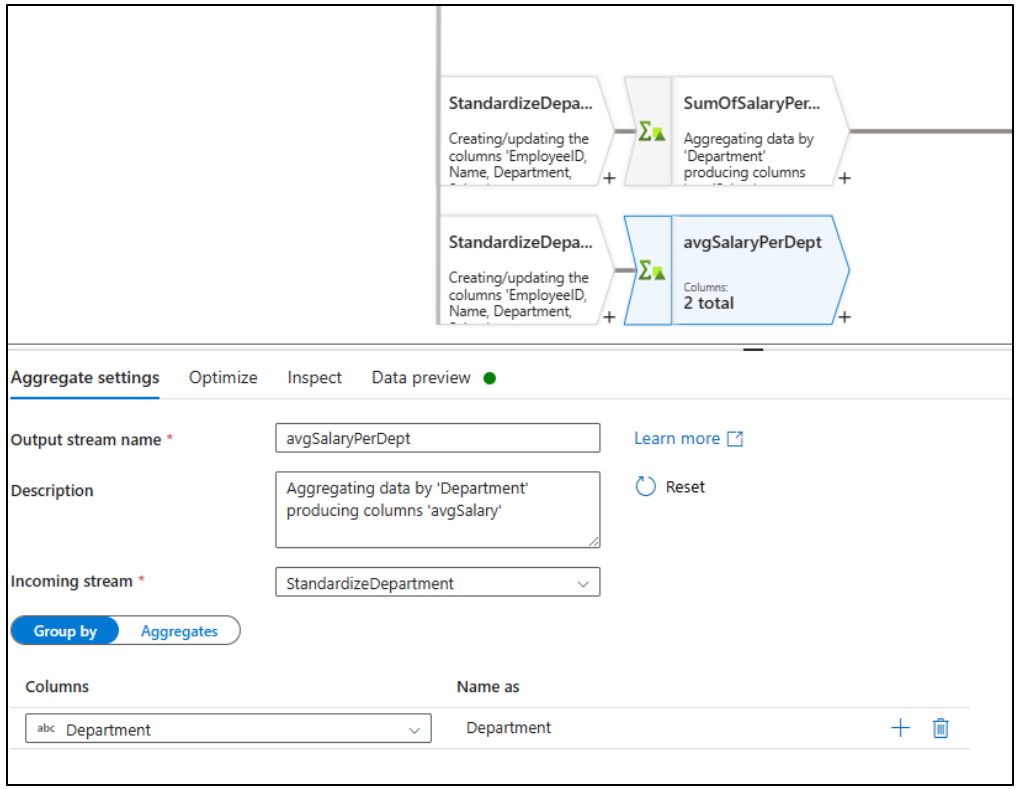


2. Data Transformation

b) Calculate the **sum of salary per department.**

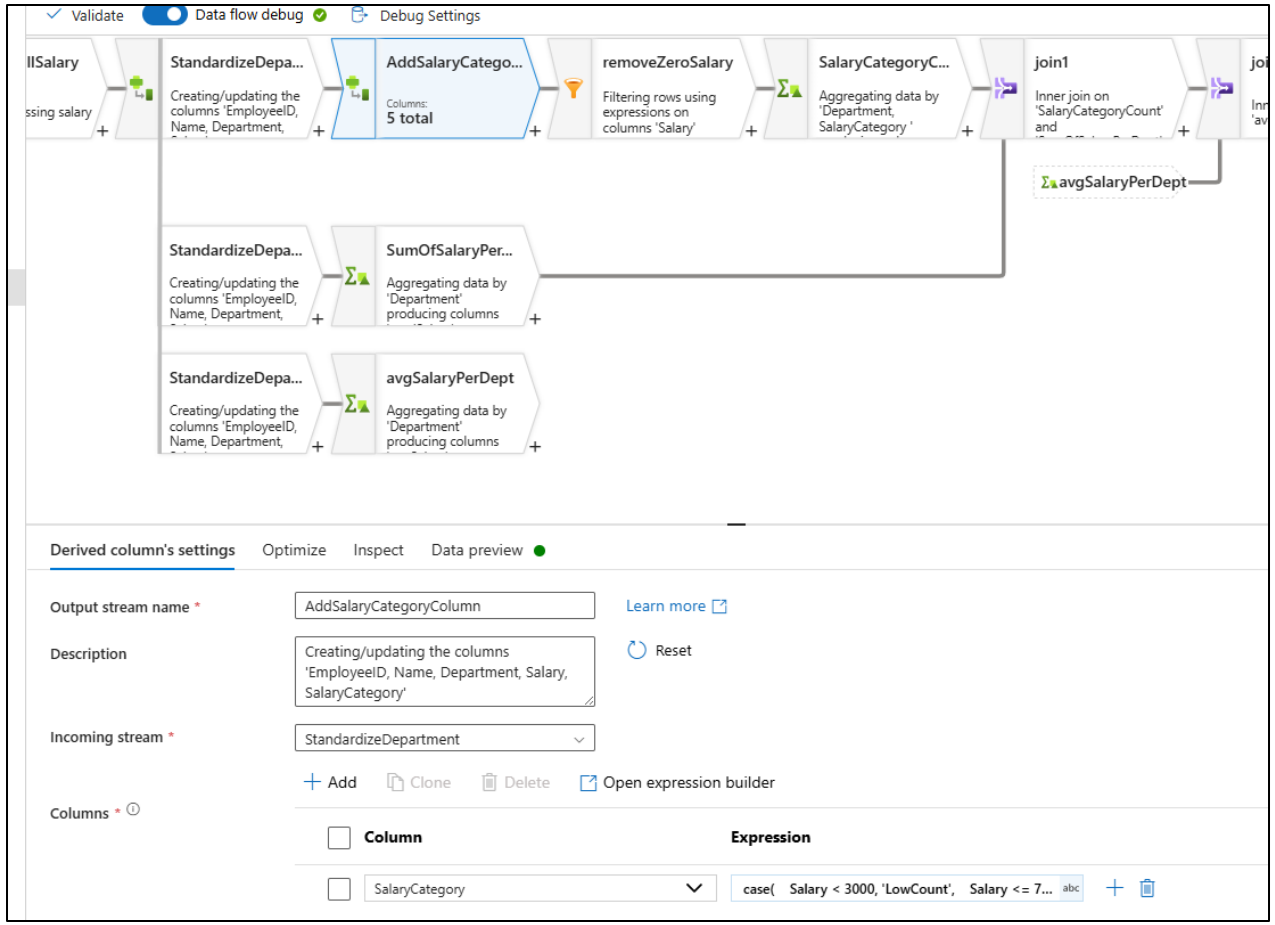
c) Calculate the **average salary per department.**





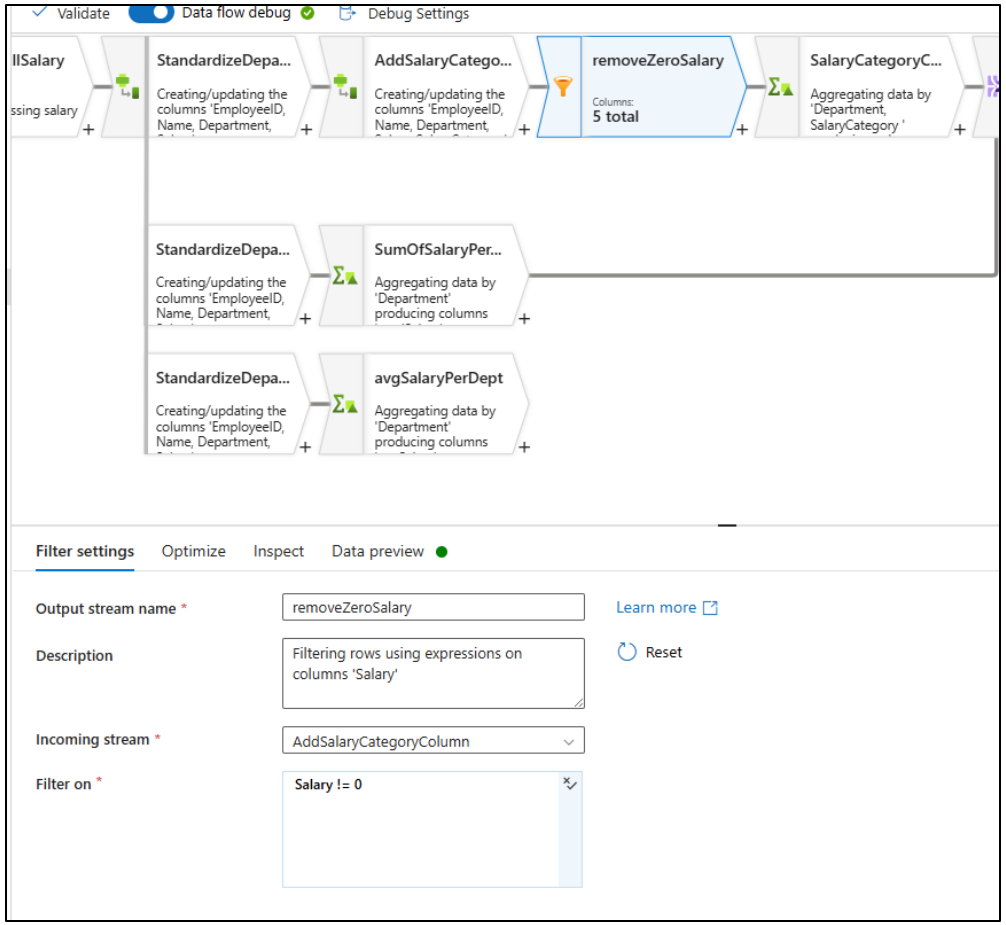
4. Derived Columns

a) Add a new column **SalaryCategory** based on salary ranges (e.g., <3000 = **LowCount**, 3000–7000 = **MediumCount**, >7000 = **HighCount**).

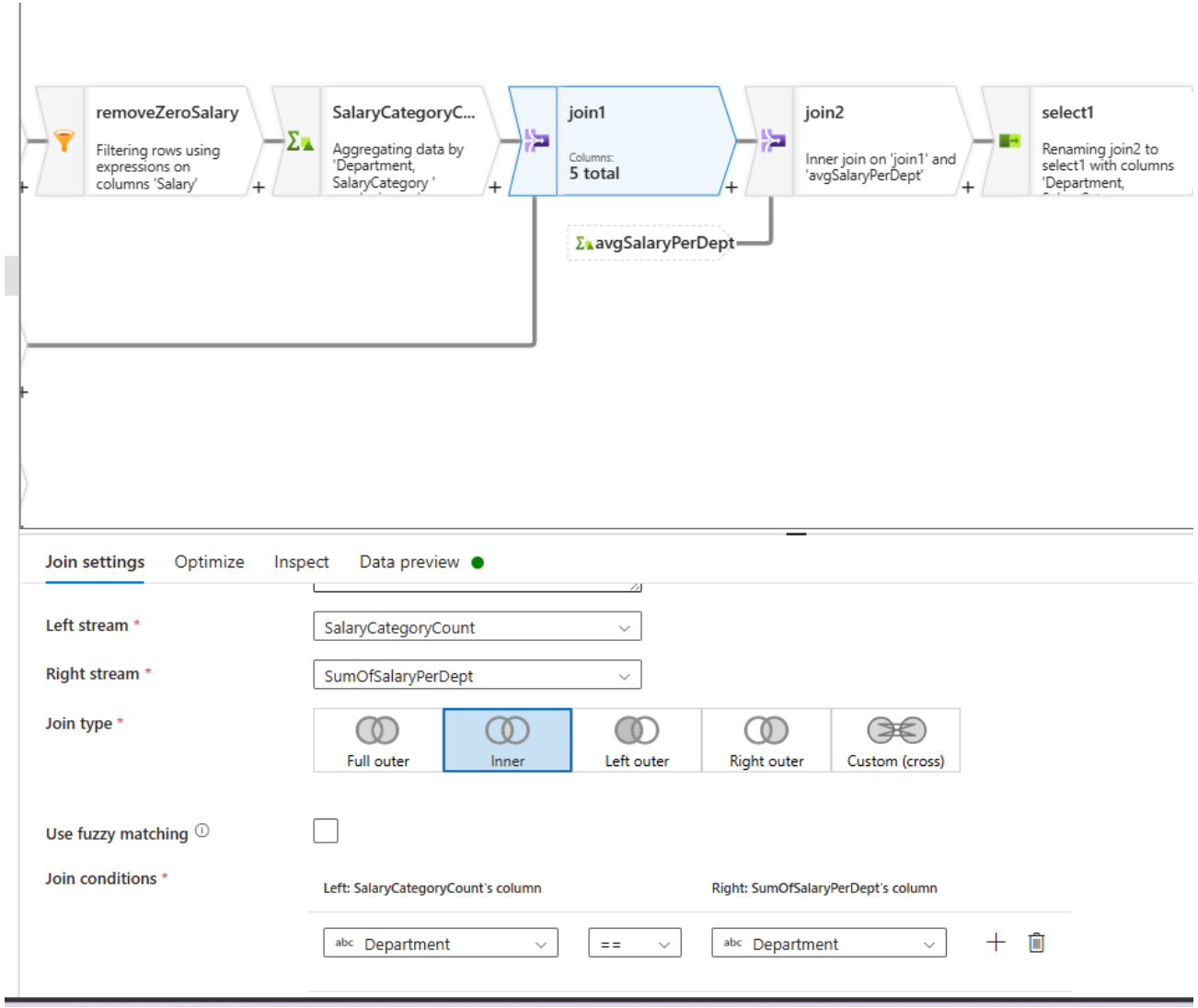


4. Filtering

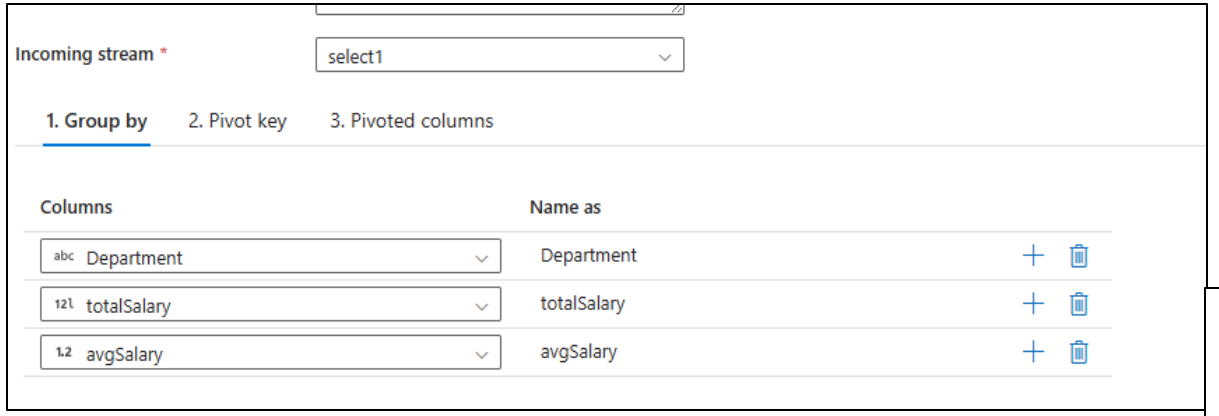
a) Exclude employees with salary = 0 after cleaning.

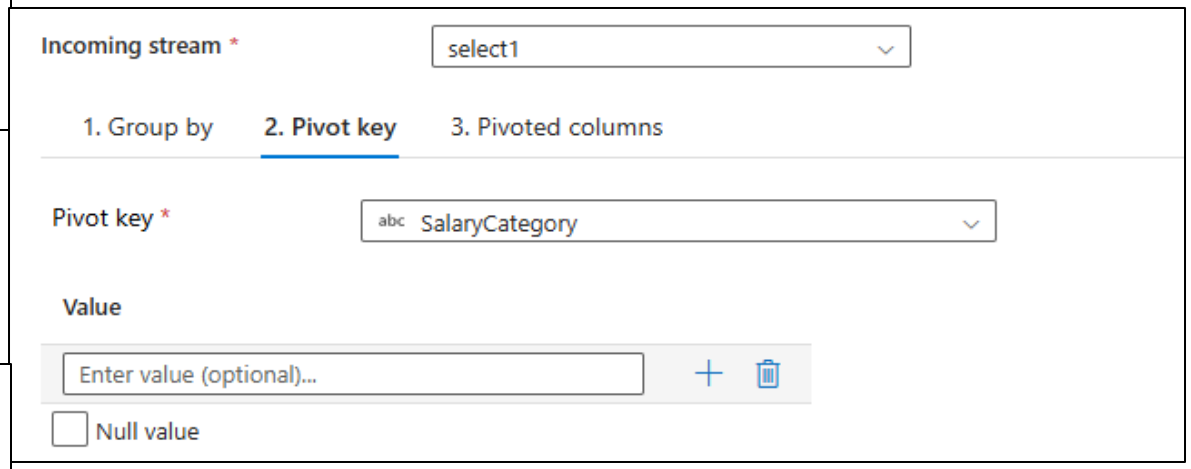


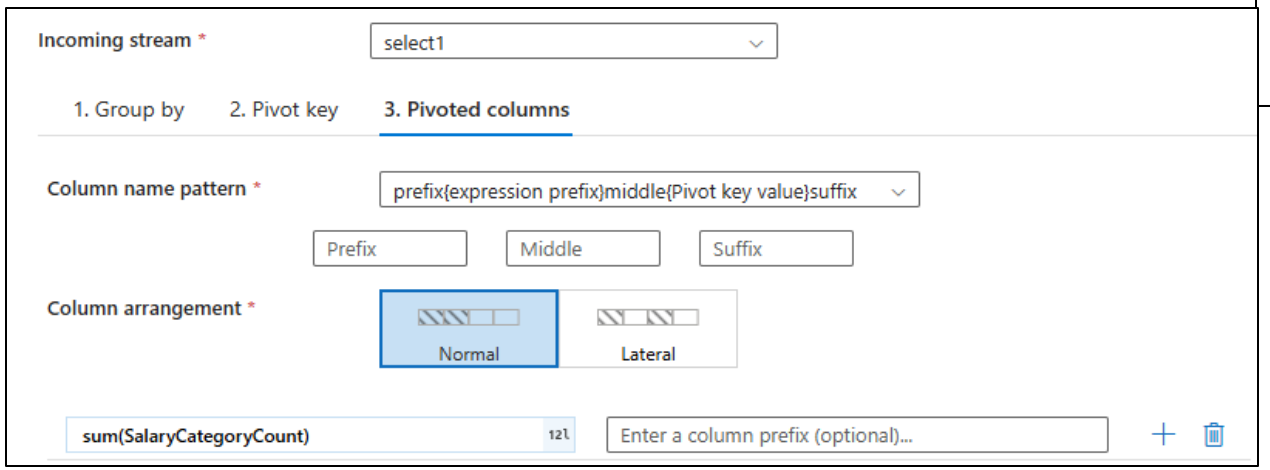
Joining main stream with total and avg salary



Pivoting the result:







Final transformed data

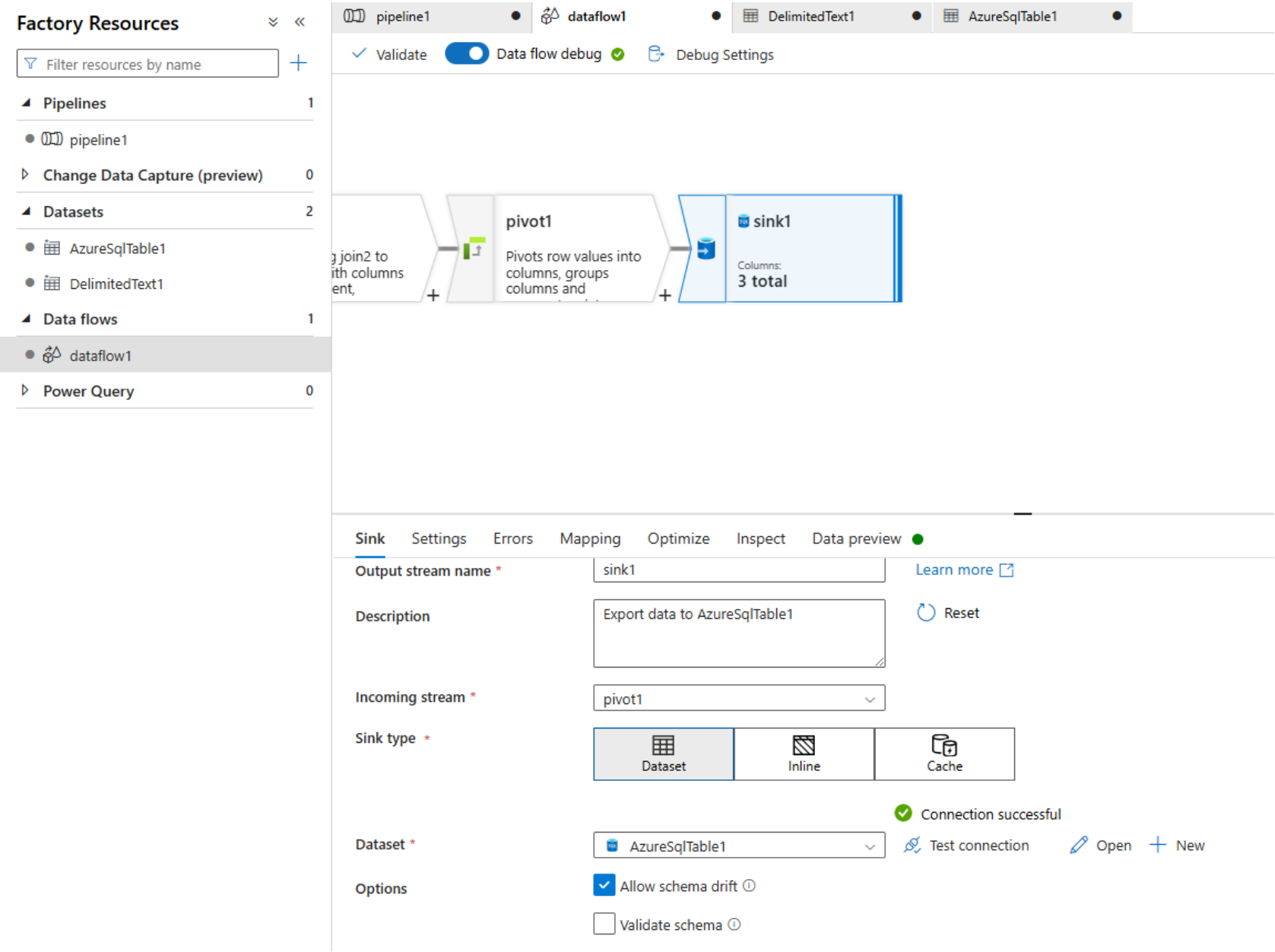
Pivot settings | Optimize | Inspect | Data preview

Number of rows INSERT 6 | UPDATE 0 | DELETE 0 | UPSERT 0 | LOOKUP 0 | ERROR 0

Refresh | Typecast | Modify | Map drifted | Statistics | Remove | Export to CSV

Department	totalSalary	avgSalary	HighCount	LowCount	MediumCount
OPERATIONS	889200	5169.767441860465	68	14	40
HR	645300	4676.086956521739	48	10	32
SALES	790900	4822.5609756097565	53	13	50
FINANCE	766100	4818.238993710692	55	12	38
MARKETING	774300	4450.0	57	8	46
IT	731100	5112.5874125874125	49	8	43

sink


Factory Resources
Filter resources by name
Pipelines 1
pipeline1
Change Data Capture (preview) 0
Datasets 2
AzureSqlTable1
DelimitedText1
Data flows 1
dataflow1
Power Query 0
pipeline1
dataflow1
DelimitedText1
AzureSqlTable1
Validate
Data flow debug
Debug Settings
pivot1
Pivots row values into columns, groups columns and
sink1
Columns:
3 total
Sink
Settings
Errors
Mapping
Optimize
Inspect
Data preview
Output stream name *
sink1
Learn more
Description
Export data to AzureSqlTable1
Reset
Incoming stream *
pivot1
Sink type *
Dataset
Inline
Cache
Connection successful
Dataset *
AzureSqlTable1
Test connection
Open
New
Options
Allow schema drift
Validate schema

Sql output

```sql
SELECT * FROM DepartmentSalarySummary;
```

Messages | Results

	Department	totalSalary	avgSalary	HighCount	LowCount	MediumCount
1	FINANCE	766100	4818.24	55	12	38
2	OPERATIONS	889200	5169.77	68	14	40
3	HR	645300	4676.09	48	10	32
4	SALES	790900	4822.56	53	13	50
5	MARKETING	774300	4450.00	57	8	46
6	IT	731100	5112.59	49	8	43

Final data-flow